Arista Access Node IND-ACC-16 Routing Protocol Flap SOP for Indiranagar Exchange

Node: IND-ACC-16 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_3049

Description

Routing protocol flaps on the Arista Access Node IND-ACC-16 at the Indiranagar Exchange can occur due to the inherent nature of dynamic routing protocols such as BGP, OSPF, and ISIS. These protocols continuously exchange routing information between neighboring nodes to build and maintain a routing table. However, in certain scenarios, this information exchange can lead to instability and oscillations, causing the routing protocol to flap. This can be attributed to factors such as suboptimal configuration, misconfigured neighbors, or transient network conditions. On Arista equipment, this is further exacerbated by the implementation of route refresh timers and the interaction with other network protocols. To mitigate this issue, it is essential to identify the root cause of the flapping and take corrective actions to stabilize the routing protocol.

Pre-requisite Checks

- check: Perform a site visit to the Indiranagar Exchange to confirm that the issue persists.

- check: Verify the status of redundant links to the Access Node IND-ACC-16.

Potential Root Causes

Cause	Probability
Misconfigured BGP neighbor	High
Inconsistent OSPF neighbor configurations	Medium
ISIS LSP flooding due to transient network conditions	Low

Diagnostic Commands & Expected Outputs

```
Command: show ip bgp summary, Expected Output: BGP table Version 1, neighbors 4, best path, table version 1, Description: This command displays a summary of the BGP table, including the number of neighbors and the best path. It helps to identify any issues with the BGP neighbor configurations.

Command: show ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface, Description: This command displays a list of OSPF neighbors, their priority, state, and interface information. It helps to identify any inconsistencies in the neighbor configurations.

Command: show isis database, Expected Output: ISIS Level 2 Database, Description: This command displays the ISIS database, including the LSPs and their associated information. It helps to identify any issues with ISIS LSP flooding.
```

Resolution Steps

Step 1: Step: Verify and correct BGP neighbor configurations, Command: show ip bgp summary | exclude best path, Description: Verify that the BGP neighbor configurations are correct and consistent. If necessary, correct the configurations to resolve the issue.

Step 2: Step: Verify and correct OSPF neighbor configurations, Command: show ospf neighbor | exclude State down, Description: Verify that the OSPF neighbor configurations are correct and consistent. If necessary, correct the configurations to resolve the issue.

Step 3: Step: Verify and adjust ISIS LSP flooding parameters, Command: show isis database | exclude LSP flooding, Description: Verify that the ISIS LSP flooding parameters are set correctly. If necessary, adjust the parameters to resolve the issue.

Post-Check Verification

```
Command: show ip bgp summary, Expected Output: BGP table Version 1, neighbors 4, best path, table version 1, Description: Verify that the BGP table is stable and the best path is correct.

Command: show ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface, Description: Verify that the OSPF neighbors are stable and their interface information is correct.

Command: show isis database, Expected Output: ISIS Level 2 Database, Description: Verify that the ISIS database is stable and there is no LSP flooding.
```

Roll-back Procedures

- Step: Restore the original BGP neighbor configuration, Command: config replace , Description: Restore the original BGP neighbor configuration from the backup configuration file.

- Step: Restore the original OSPF neighbor configuration, Command: config replace , Description: Restore the original OSPF neighbor configuration from the backup configuration file.

- Step: Restore the original ISIS LSP flooding parameters, Command: config replace , Description: Restore the original ISIS LSP flooding parameters from the backup configuration file.